Exercises

Algebraic problems

For this section we are concerned with **homogeneous, linear, second-order ODEs** which resemble the following expression:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

To find a general solution for y we first write out the **auxiliary** equation

$$a\lambda^2 + b\lambda + c = 0$$

and solve it to find the roots. If the roots are real and not equal ($\lambda_1 \neq \lambda_2$ then we can write a general solution as:

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

If our two roots are real and equal (i.e. $\lambda_1 = \lambda_2$) then we use the formula:

$$y = (A + Bx)e^{\lambda_1 x}$$

We don't consider what happens when the two roots are not real (i.e. $b^2 - 4ac < 0$) in this course.

Question 9.1: For the following second-order ordinary differential equations, write out the auxiliary questions, find the roots and express the general solution:

(a) $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} - 16y = 0$
(b) $8\frac{d^2y}{dt^2} - 2\frac{dy}{dt} - y = 0$
(c) $4\frac{d^2y}{dt^2} - 24\frac{dy}{dt} + 27y = 0$
(e) $\frac{d^2y}{dt^2} - 4y = 0$

Question 9.2: Return to the general solutions found in Question 9.1 and use the following initial or boundary conditions to find the constants A and B and then state the full solution.

(a) $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} - 16y = 0$ using $y = 2$ and $\frac{dy}{dx} = -6$ when $x = 0$.
(b) $8\frac{d^2y}{dt^2} - 2\frac{dy}{dt} - y = 0$ using $y = 10$ and $\frac{dy}{dx} = 2$ when $x = 0$.
(c) $4\frac{d^2y}{dt^2} - 24\frac{dy}{dt} + 27y = 0$ using $y = 0$ and $\frac{dy}{dx} = 12$ when $x = 0$.
(d) $\frac{d^2y}{dt^2} - 4y = 0$ using $y = -4$ and $\frac{dy}{dx} = 0$ when $x = 0$.

Question 9.3: Solve the following differential equation given the associated initial conditions associated with both the function value for y and gradient $\frac{dy}{dx}$.

$$3\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 8y = 0$$

where

$$y = -6 \text{ and } \frac{dy}{dx} = -18 \text{ when } x = 0$$

Question 9.4: For the following second-order ordinary differential equations, write out the auxiliary questions, find the roots and express the general solution:

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

Real-world problems

Question 9.5: Newtons laws of motion (as 2nd Order ODEs): We can write Newtons second law of motion as follows:

$$\begin{aligned} F &= m\frac{dv}{dt} \\ F &= m\frac{d^2x}{dt^2} \end{aligned}$$

(a) Use Newton's second law of motion to derive the equation of motion (distance versus time) of an object moving down a ramp inclined at 30°, starting with velocity $v = 0$. Do not consider air resistance or friction.

(b) Use Newton's second law of motion to derive the equation of motion (distance versus time) of an object moving down a ramp inclined at 30°, starting with velocity, $v = 0$. Do not consider air resistance but consider the frictional force to be equal to $\mu \times N$, where N is the magnitude of the contact force and $\mu = 0.1$. Solve this for initial velocity $= 0$ to get the position x vs time.

Question 9.6: Fick's Law of diffusion (as 2nd Order ODEs): An industrial incinerator produces a steady output of particulate matter having a concentration of $C_0 = 1.0$ mg m^{-3} of air at the chimney. 1 km away from the incinerator plant the concentration is $C_{1km} = 0.1$ mg m^{-3}. Some houses are a distance of 500 m away from the plant. Using Fick's law of 1-d diffusion, $\frac{d^2C}{dx^2} = 0$, what is the concentration of pollutant where the houses are? Note that you have been told the limit for particulate matter is 0.5 mg m^{-3}. Is the concentration at the location of the houses above 0.5 mg m^{-3}?

Question 9.7: Fouriers law of heat transfer (as 2nd Order ODEs): 3 km below the surface there is a hot granite bed, with a steady-state temperature of 250 °C. The temperature just below the surface the temperature stays relatively constant at 10°C, use the steady-state Fouriers law in 1-d, $\frac{d^2T}{dx^2} = 0$ to derive an equation that describes the temperature profile between the surface and the hot granite.

Question 9.8: Newtons laws of motion (as 2nd Order ODEs): Derive an equation for the displacement with time if an object moving at an initial velocity is experiencing a constant acceleration. In other words, we know that we can write Newtons second law as a second order ODE):

$$\frac{d}{dt}\left(\frac{dx}{dt}\right) = a$$

where a is now a constant. Proceed with integrating this expression by assuming that the starting position is $x = 0$ and initial velocity is v_0 m/s.